

PROBLEMS

1. Write the following propositions symbolically

- (i) I play cricket or I go to a movie
- (ii) 5 is an odd number and $5 + 5$ is an even number
- (iii) My father will send me abroad if and only if I get a rank in the examination
- (iv) If a quadrilateral is not a parallelogram, then it is not a rectangle.

2. p: Question papers are easy

q: We pass the exam

r: Principal declares holiday

s: we are happy

Give the verbal meaning of the following propositions

- (i) $p \rightarrow q$ (ii) $(p \vee r) \rightarrow s$

3. Let p: $3^2 + 4^2 = 5^2$, q: $\sqrt{2}$ is a rational number, r: $7 > 3$

Determine the truth values of the following propositions

- (i) $p \vee q$ (ii) $q \wedge r$ (iii) $p \leftrightarrow r$

- (iv) $\neg[(p \rightarrow q) \vee (q \rightarrow r)]$

- (v) $(p \wedge \neg q) \vee (q \wedge \neg r)$

4. Find the possible truth values of p, q and r in the following cases

- (i) $p \rightarrow (q \vee r)$ is false

- (ii) $p \wedge (q \rightarrow r)$ is true

5. Let p, q, r be propositions having truth values 0, 0, 1 respectively. Find the truth value of the following proposition.

(i) $(p \vee q) \vee r$ (ii) $p \rightarrow (q \wedge r)$ (iii) $p \rightarrow (q \rightarrow \neg r)$

(iv) $p \wedge (q \rightarrow r)$

6. If a proposition q has the truth value 1, determine all truth value assignments for the primitive propositions p, r and s for which the truth value of the following compound proposition is 1

$$[q \rightarrow \{(\neg p \vee r) \wedge \neg s\}] \wedge [\neg s \rightarrow (\neg r \wedge q)]$$

7. Construct the truth tables for the following

(i) $\neg(p \rightarrow q)$ (ii) $(p \vee q) \rightarrow r$

8. Determine whether the following compound propositions are tautology, contradictions or neither

(i) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$

(ii) $(p \vee q) \wedge (\neg p \wedge \neg q)$

(iii) $p \rightarrow (\neg p \vee q)$

(iv) $p \vee (q \wedge r) \leftrightarrow [(p \vee q) \wedge (p \vee r)]$